Measurement, moderation and missingness in twin models

A partial replication of Schwabe (2016)

Abstract

We partly replicated Schwabe's (2016) finding that psychometric modelling improves behavioural-genetic inference (Schwabe, 2016): all 108 reconstructed precision contrasts favour full information (illustrative published gain 0.00106356; ours 0.00150235). Whole-dissertation replication is incomplete: empirical refits, repeated-dataset Bayesian recovery and direct power are not established. Eleven full-information bias signs reverse the source; none excludes zero after adjustment. One published heritability bias is 0.03315402 versus our 0.03118039. Source precision magnitudes remain unresolved despite agreeing directions; conditional mathematics and one completed posterior diagnostic do not establish broad recovery or likelihood uniqueness.

Scientific estimand / claim	Published finding	Our finding	Effect magnitude	Direction	Verdict
HE01 Subsets [N]	Item-subset invariance	Conditional population support	not established	not established	partial
HE02 IRT recovery [N]	IRT remedies scale bias	Aggregate recovery incomplete	not established	not established	unresolved
HE03 Power [N]	Useful detection power	Detection rates not established	not established	not established	unresolved
HE04 Sum scores [N]	Spurious interactions	Single-case diagnostics only	not established	not established	unresolved
HE05 Math AxE [S]	2.0837	not established	not established	not established	blocked
HE06 Math AxC [S]	2.6159	not established	not established	not established	blocked
HE07 Dimensions [N]	Two response dimensions	Original inputs unavailable	not established	not established	blocked
HE08 Subscale [N]	Nine-item subscale	Original inputs unavailable	not established	not established	blocked
HE09 Liberalism AxE [S]	-2.81	not established	not established	not established	blocked
HE10 Liberalism AxC [S]	0.54	not established	not established	not established	blocked
HE11 AxE reversal [N]	AxE direction reverses	Original inputs unavailable	not established	not established	blocked
HE12 AxC reversal [N]	AxC direction reverses	Original inputs unavailable	not established	not established	blocked
HE13 Identification [N]	Uniquely identified model	Conditional identification only	not established	not established	partial
HE14 Bias prevention [N]	Prevents measurement bias	General prevention untested	not established	not established	unresolved
HE15 One maximum [N]	One likelihood maximum	Finite-maximum uniqueness unresolved	not established	not established	unresolved
HE16 A moderation [N]	Large samples needed	Direct power untested	not established	not established	unresolved
HE17 C moderation [N]	Large samples needed	Direct power untested	not established	not established	unresolved
HE18 20 items [N]	Twenty items sufficient	Sufficiency untested	not established	not established	unresolved
HE19 SES C slope [S]	-3.0777	not established	not established	not established	blocked
HE20 Nondetection [N]	13 effects not detected	Original inputs unavailable	not established	not established	blocked
HE21 Power gain [N]	FI more powerful	Direct power untested	not established	not established	unresolved
HE22 Precision [N]	FI gives lower variance SE	108 positive SE contrasts	not established	not established	partial
HE23 Likelihood [N]	All observed components	Gaussian construction verified	not established	not established	partial
HE24 MAR A bias [N]	Deletion A low; FI close to truth	Sign agrees; closeness untested	not established	not established	partial
HE25 MAR C bias [N]	Deletion C low; FI close to truth	Sign agrees; closeness untested	not established	not established	partial
HE26 MAR h2 bias [N]	Deletion heritability high; FI close to truth	Sign agrees; closeness untested	not established	not established	partial
HE27 A SE gain [S]	0.00106356	0.00150235	not established	same direction	partial
HE28 h2 bias [S]	0.03315402	0.03118039	smaller	same direction	point only

N denotes a nonscalar claim; S a scalar effect; FI full information. Descriptive point estimates; not equivalence. Nonscalar rows have no pooled effect or numeric margin; their scientific verdict is separate. Scalar slopes are in log-variance units; rounding tolerances: HE05–06,19, 0.00005; HE09–10, 0.005. HE27 is deletion minus full-information variance SE (squared phenotype units), positive favouring full information, tolerance 0.00023667095377570213. HE28 is estimate minus generating heritability (proportion), positive indicating upward bias, tolerance 0.00027219389842495325. These two margins bound source-figure extraction, not source Monte Carlo error. Definitions, source locators and all component comparisons follow.

1 Introduction

We partly replicated Schwabe's (2016) finding that psychometric modelling improves behavioural-genetic inference: the reconstructed missing-covariate experiment supports the full-information precision ordering across all 108 comparisons, but does not establish direct power, empirical applications, or the complete Bayesian recovery claims. The published and reconstructed deletion-minus-full-information standard-error gains in the first illustrative cell are 0.00106356 and 0.00150235; the corresponding source and reconstructed heritability biases in the second illustration are 0.03315402 and 0.03118039. Source-figure extraction margins are respectively 0.00023667095377570213 and 0.00027219389842495325. The first magnitude is unresolved because the original standard-error implementation is unknown; the second point magnitude is smaller. Neither comparison is an equivalence test. Eleven full-information bias estimates reverse the source point sign, with no adjusted interval excluding zero.

The scientific question is whether modelling measurement and missingness changes conclusions about genetic and environmental variation. The 2016 University of Twente dissertation, *Nature, Nurture and Item Response Theory: A Psychometric Approach to Behaviour Genetics*, combines item response theory (IRT), twin variance models, simulations, empirical applications and software (Schwabe, 2016). IRT models the probability of each item response given a latent trait. The twin model separates additive genetic variance (A), shared environmental variance (C) and individual environmental variance (E); monozygotic and dizygotic genetic correlations are fixed at 1 and 1/2. These model assumptions matter as much as numerical agreement.

The unit of scientific replication is a substantive proposition and its design, not an inventory row. Our coverage includes all seven chapters and Appendices A–G, with 996 linked original-claim records, 26 principal claims, two preselected numerical illustrations and 187 retained headline component comparisons. Parents share their constituent datasets, estimates and assumptions. The records therefore define neither 996 independent findings nor a replication success fraction. Translated summaries and repeated conclusions map to their original chapters; cited background and future proposals are not new completed experiments.

2 What we did ourselves, and what we took from others

We reconstructed the missing-data experiment, checked mathematical and software semantics, generated response datasets and evaluated bounded posterior implementations. The dissertation supplied the models, designs, source tables and figures. Our work supplied original reconstruction code, generated data, fitted values, analytical arguments, uncertainty calculations and explicitly stated choices where source details were absent. Reading a number from a figure or evaluating a formula using printed parameters is source-based arithmetic, not an independent empirical refit.

The primary Gaussian experiment used 1,000 datasets in each of 36 conditions and 12 additional sensitivity conditions. Full-information and deletion fits on the same dataset give 96,000 method fits: 72,000 primary and 24,000 sensitivity fits. All completed without a failed or guard-invalid fit in the saved analysis. Separately, response generation produced 2,650 datasets across 19 Chapter 2/5 conditions; paired reliability sensitivity reused those datasets and fitted no posteriors. A four-chain posterior diagnostic concerns only one fixed Chapter 2 dataset. The bounded score-likelihood implementation uses that same observed dataset and adds no independent dataset.

The primary outcomes and exact design accounting are in `production/chapter6-v1/terminal-analysis/summary.json`, `production/chapter6-v1/terminal-analysis/condition-summary.csv` and `production/analysis-plan.json`. The complete source-linked inventory is `coverage-v4.csv`; `handoff/a0-evidence-packet.json` preserves each claim's published quantity, our evidence, uncertainty,

source locator, dependencies and missing inputs. The auxiliary feasibility outcome leaves the original claim verdicts unchanged, as specified in `provenance/score-a0-coverage-status.json`.

3 Missing covariates: precision, bias and power

Full information improves precision in the reconstructed Chapter 6 design, while the source's broader power assertion is not established. The primary grid crosses missing completely at random (MCAR), missing at random (MAR) and missing not at random (MNAR), two through five covariates, and missing fractions 0.02, 0.06 and 0.10. An additional 0.03 grid addresses the source's conflict between 2% prose and 3% plot labels. It does not silently replace the primary design. The comparison concerns residual ACE components, not phenotype variance inclusive of covariate explanation.

3.1 The precision ordering is reproduced, its literal magnitude is unresolved

All 108 signed variance-standard-error contrasts, deletion minus full information, are positive. Each of the 108 Bonferroni-adjusted Student-t Monte Carlo intervals also lies above zero. Pairing preserves the fact that both methods fit the same generated dataset; the Monte Carlo standard error (MCSE) is the standard deviation of replicate differences divided by the square root of the replicate count. Familywise adjustment is over the 108 primary contrasts, using the Bonferroni principle (Dunn, 1961). This is evidence about the reconstructed grid's mean precision ordering.

All 216 literal plotted variance-standard-error points fall outside the conditional source-comparison bands. That diagnostic disagreement cannot by itself establish failure to reproduce the original estimand: original standard-error implementation and source Monte Carlo uncertainty are unavailable. A path standard error concerns a different parameter from a variance standard error. Its exploratory agreement must not replace the dissertation's stated variance estimand. The full signed values and conditional comparison records are retained in `production/chapter6-v1/terminal-analysis/signed-SE-contrasts.csv` and `production/chapter6-v1/terminal-analysis/source-comparisons.csv`.

For MCAR, two covariates and 2% missingness, the source A-variance SE gain is 0.0010635603 and ours is 0.0015023487. Our paired MCSE is 0.0000223093; the 108-family adjusted interval is [0.0014239767, 0.0015807208]. Both point signs favour full information. The geometric source margin is 0.0002366710, but source-estimand comparability is unresolved, so the magnitude label remains not established. The conditional point difference of 0.0004387884 is not a new scientific contrast or evidence of equivalence.

3.2 MAR bias directions agree without establishing full-information unbiasedness

Deletion underestimates A and C and overestimates residual heritability in all 12 MAR cells for each component. Adjusted intervals exclude zero in the stated direction in 10 A cells, all 12 C cells and nine heritability cells. The point ranges are respectively [-0.07283115, -0.00621756], [-0.11495778, -0.00766858] and [0.00141622, 0.03118039]. The complete 36-cell directional finding is not 36 independent discoveries.

Full-information point biases are much nearer zero in this reconstruction, with ranges [-0.00319176, 0.00133589] for A, [-0.00261639, 0.00101715] for C and [-0.00139029, 0.00265364] for heritability. None of their adjusted intervals excludes zero. This does not establish zero bias or the source's qualitative claim of closeness: no scientific equivalence margin was specified. Eleven full-information point signs reverse their source signs; none of those reversals is supported by an adjusted exclusion of zero. Two conditional MAR deviations from source estimates occur at the ambiguous low missingness level.

For deletion with five covariates and 10% MAR missingness, the generating residual heritability is 0.5.

The source mean 0.53315402 gives signed bias 0.03315402; our mean 0.53118039 gives bias 0.03118039. Our point magnitude is smaller at the geometric margin 0.0002721939, and the direction is the same. Our MCSE is 0.00304897 and the exploratory 72-family adjusted interval is [0.02080654, 0.04155423]. The magnitude label is descriptive, not a test of a difference from the source. All 72 signed MAR bias comparisons are retained in production/chapter6-v1/terminal-analysis/signed-bias-contrasts.csv.

3.3 Additional trends do not settle power or adequacy

All 81 adjacent-covariate-count deletion variance-SE point differences in the saved assessment are positive. Absolute C bias exceeds absolute A bias in all 12 primary MAR cells, but no new cross-parameter interval or test was run. With 10% MNAR missingness, full-information E means for two through five covariates are 0.20821236, 0.20733095, 0.20619935 and 0.20476346. The last is nearer 0.20 than 0.21, unlike the first three. The source's per-condition outputs are unavailable.

Twenty-one of 26 saved targets have complete descriptive assessments. Direct MCAR, MAR and MNAR power targets, unequal zygosity structure, and the broad power assertion remain pending. The recommendation-relevant subset has 54 positive adjusted precision contrasts out of 54, but this does not establish an optimal threshold of more than 3% missingness and more than two covariates. A power comparison needs a specified nonzero alternative, test and rejection rule, followed by repeated-dataset rejection rates. Precision alone does not supply it. The signed assessment is outputs/chapter6-saved-target-assessment.json.

4 Measurement scales and mathematical identification

Conditional population results support parts of the dissertation's mathematical argument, but do not establish universal item-subset invariance or a unique finite-sample likelihood maximum. These are ordinary mathematical derivations, not proof-assistant certificates. They neither verify source empirical assumptions nor supply missing posterior experiments.

4.1 A common scale identifies a target, not identical sample evidence

With known positive logistic calibrations, distinct locally independent item observations, positive Gaussian ACE components and both zygosities at two shared moderator levels, admissible item subsets identify the same population parameter on the common anchored scale. Within-person marginal and joint probabilities identify the mean and total latent variance. Cross-twin response probabilities then identify latent covariances, because their Gaussian covariance derivative is positive for positive-slope response functions. The covariance derivative identity is a form of Price's theorem; the evidence uses its bounded smooth-function setting (Voigtlaender, 2020). Recover

$$A = 2(K_{\mathrm{MZ}} - K_{\mathrm{DZ}}), \quad C = 2K_{\mathrm{DZ}} - K_{\mathrm{MZ}}, \quad E = T - K_{\mathrm{MZ}}.$$

Two moderator values identify the log-linear variance and affine mean coefficients under the stated common functions. Unknown calibration or unobserved joint response moments break this argument.

Equal population targets do not imply equal likelihood evidence in a finite sample. An exact example holds total variance at 1 and compares $(A, C, E) = (1/4, 1/4, 1/2)$ with $(1/2, 1/8, 3/8)$. DZ covariance is 3/8 at both points while MZ covariance rises from 1/2 to 5/8. Two admissible Rasch subsets with different observed concordance patterns yield opposite likelihood orderings of these parameter points. This establishes neither opposite unrestricted maximum-likelihood estimates nor opposite significance

decisions. The dissertation itself qualifies scaling by common identification and acknowledges weak tail information (pp. 51, 107–108). See handoff/scale-invariance-evidence.json.

4.2 Log-variance moderation removes one symmetry, not all inferential difficulties

For a shared binary moderator $m \in \{0, 1\}$ and path $a + bm$ with $a > 0$, the reflected slope $b^* = -2a - b$ leaves both endpoint squared paths unchanged. The alternatives coalesce if $a+b = 0$. With three distinct moderator values, equality of two squared affine paths implies equality up to a global sign; strict positive intercepts remove that global ambiguity. Thus the binary reflection need not extend to rich continuous support. Unrestricted global sign symmetry does extend. These are distinct interpretations of Appendix C.

Log-linear variance $V(m) = \exp(\beta_0 + \beta_1 m)$ has derivative $\beta_1 V(m)$, establishing monotonicity for a fixed slope. It does not establish likelihood concavity. The Gaussian model with common component functions has within-twin variance $T = A + C + E$ and cross-twin covariance $A + C$ for MZ and $A/2 + C$ for DZ. Source square-root scaling in Eq. 5.14 is essential to this calculation. The verified moments and symmetry arguments are in handoff/chapter5-analytical-evidence.json.

A separate sufficient identification condition uses common exact log-linear component functions and affine mean functions over moderator laws that, within each zygosity, are not concentrated on three or fewer values. Four distinct atoms suffice, and the two zygosity supports may be disjoint. It does not justify arbitrary smooth extrapolation across disjoint supports, binary supports, noisy separate-twin moderators or unknown calibration. Failure to satisfy a sufficient condition is not proof of nonidentification. The explicit support assumptions and result are retained in provenance/continuous-moderator-identification-check.json.

4.3 Population identification does not prove one finite likelihood maximum

The source links unique identification to one likelihood maximum (Chapter 5, p. 72). Under a free common mean, an all-zero response sample has likelihood increasing toward its supremum as that mean decreases, without attaining a finite maximum. This refutes universal finite-maximum existence, a stronger interpretation than at-most-one maximum when one exists. It is not a multiple-maximum example. Neither a proof nor a counterexample for the general at-most-one finite local or global maximum claim is supplied.

The Chapter 2 genetic moderation model also requires a distinct observation argument. Nonzero interaction gives non-Gaussian marginal traits. A conditional fixed-parameter latent-moment inversion can identify latent components under known genetic correlations, common component laws and independent innovations. It does not show that finitely many calibrated Rasch response patterns identify those latent moments. The Gaussian item argument cannot supply that missing bridge. Fixed-parameter moment existence also does not imply finite prior-predictive moments under inverse-gamma$(1, 1)$ variance priors. See provenance/non-gaussian-latent-moment-check.json.

5 Response generation and Bayesian feasibility

The reconstructed generators and one completed posterior diagnostic provide limited evidence about feasibility, not the dissertation's repeated-dataset recovery or power claims. Chapter 2's 100-dataset and Chapter 5's 250-dataset posterior aggregates remain incomplete. Generation counts, retained draws and completed source-target experiments are different quantities.

5.1 Generation matches some descriptors and leaves the source estimand uncertain

The response experiment comprises 14 Chapter 2 cells with 100 datasets each and five Chapter 5 cells with 250 datasets each. The seed rule is $910000 + 1000c + r$, with condition c and replicate r fixed before generation. The initial pilot duplicates each cell's first seed and is not 19 extra observations. All 19 conditions, including unfavourable descriptor comparisons, are reported in handoff/generator-summary-scientific-evidence.json.

For Chapter 2 floor and ceiling designs, source reliability is approximately 0.90, whereas our raw-alpha means are 0.90802950 and 0.90753316 (MCSEs 0.00059652 and 0.00050631). All 12 power-design means round to the reported 0.92. Chapter 5's four 60-item cells round to 0.90, while the 20-item mean 0.73819631 rounds to 0.74 rather than the quoted 0.75. Source aggregation, calibration and participant subset are not determined sufficiently to turn these descriptors into exact-source verdicts.

For item covariance V, raw alpha is $J[1 - \mathrm{tr}(V)/\sum V]/(J-1)$; standardized alpha is $J\bar{r}/[1 + (J-1)\bar{r}]$. Reusing the same 2,650 generated datasets gives floor and ceiling standardized means 0.90697262 and 0.90645093. For 20 items, standardization lowers 0.73819631 to 0.73109750, a paired change of -0.00709880 with MCSE 0.00010938. This does not explain the descriptor gap. The paired calculation adds zero independent datasets and zero posterior fits; see handoff/generator-alpha-sensitivity-scientific-evidence.json.

The source floor skewness endpoints 0.630 and 0.632 and ceiling endpoints -0.435 and -0.434 refer to different estimators on an unknown illustrative sample. They are not across-replicate extrema. Our replicate skewness means and observed extrema cannot be substituted for those endpoints. Reconstructed positive floor and negative ceiling skew support qualitative directions, not the exact figure or an undefined qualification such as slight.

5.2 One completed centered diagnostic supports only its fixed case

The completed centered item-Bernoulli model uses one $N = 500$, $J = 60$ floor dataset, seed 301, unchanged priors and known calibration. Four chains each have 1,000 adaptation, 7,000 burn-in and 28,000 retained iterations. All 14 monitored quantities pass the declared joint operational criteria: rank-normalized split/folded $\widehat{R} < 1.01$, bulk and tail effective sample sizes at least 400, and mean MCSE/posterior SD at most 0.05. These diagnostics follow the rank-based approach of Vehtari et al. (2021); they are finite diagnostics, not a proof of convergence.

The worst $\widehat{R}$ is 1.001673, minimum bulk effective sample size 685.762, minimum tail effective sample size 1407.532, and maximum relative mean MCSE 0.038211. Interaction mean is -0.12338265, with pooled 95% highest-posterior-density (HPD) interval [-0.68469410, 0.43069918]. All 12 whole/half-chain interaction HPDs contain zero. This is non-exclusion on one dataset, not equality to zero, unbiasedness, a calibrated detection rate or replication of the source's aggregate. Equal-tailed quantile MCSE is not HPD endpoint MCSE.

Shorter original and centered diagnostics failed their joint criteria. The first 12,000 retained draws of each completed longer chain repeat the shorter centered run exactly; those prefixes must not be pooled or counted as independent evidence. The independently checked result still shares data, model, backend and some diagnostic-library computations. Variance components retain serial correlation and rank differences. The full summaries and limitations are in handoff/jags-centered-extended-recovery-scientific-evidence.json.

5.3 An exact score representation did not complete the fixed workload

For known Rasch difficulties, the likelihood of a response vector depends on the latent trait through its canonical score, up to a known parameter-independent factor. The exact calibrated score likelihood therefore preserves the physical latent model and priors. It is not the dissertation's Gaussian analysis of sum scores. Exact target preservation does not promise computational acceleration; see `provenance/rasch-sufficiency-derivation.json`.

The bounded score-model test requested the same four-chain, 28,000-retained-draw workload on the same fixed dataset, using JAGS 4.3.2. All four chains adapted, but each retained only 16,000 draws before the predetermined summed-process budget ended: 64,000 saved draws and zero completed fixed-length chains. Chain/cleanup time was 6419.751313 seconds; all-phase summed time was 6420.894927 seconds and elapsed wall time 1606.104006 seconds. These are distinct resource measures, not CPU seconds.

The completed item-Bernoulli reference used 4927.558112 summed chain-process seconds for four chains of 28,000 retained draws. The score implementation consumed more chain/cleanup time while retaining fewer draws. This is neither equal-work benchmarking nor proof that score methods cannot accelerate other implementations: starts, random streams and runtime epochs differ. Computational acceleration and posterior agreement are not established.

The full fixed-draw joint diagnostic and five-global MCSE comparisons are ineligible and were not run. Adaptation and finite checkpoints do not substitute for convergence evidence. Incomplete chains are not pooled with historical chains. This same-backend feasibility test contributes no new dataset, no completed source-count aggregate and no change to the original scientific verdicts. Exact outcome evidence is `handoff/score-a0-scientific-evidence.json` and `provenance/score-a0-terminal-outcome.json`.

6 Empirical findings and software semantics

Independent empirical replication is blocked by missing source responses, linkage and calibration inputs. The available source reports remain scientifically important, but source arithmetic cannot replace their refitting. In Chapter 3 mathematics, the published genetic-by-individual-environment slope is 2.0837 and the genetic-by-shared-environment slope is 2.6159; the source reports the former HPD excluding zero and the latter including zero (Table 3.2, p. 42). Our independent slopes are not established.

Chapter 4 reports a two-dimensional interpretation of 28 items and a nine-item social-liberalism subscale. Its IRT AxE slope is -2.81 and AxC slope 0.54 (Table 4.3, p. 63), both with reported HPDs excluding zero. The directions change in the sum-score analysis. Comparing absolute values across differently identified scales would conceal the substantive sign reversal; subtracting those unstandardized slopes would not create a justified scalar effect. Neither dimensional interpretation, scale selection nor interaction reversal was independently refitted. The published article and supplement are distinct source materials (Schwabe et al., 2016); accessible plots and programs do not supply Virginia30K responses.

Chapter 5 reports shared-environment variance decreasing in high-SES families, with log-variance slope -3.0777 and an HPD excluding zero (Table 5.4, p. 82). Chapter 6 reports no detected coefficient among 13 measured covariates under its 99.6% HPD criterion (Table 6.3, p. 100). These application results require the original linked NTR/Cito data; the simulated Gaussian missingness experiment cannot reproduce them. Nondetection does not imply all 13 population effects are zero.

Some software propositions can be evaluated without the empirical data. For positive-slope logistic items, conditional information is $a_j^2 p_j(1-p_j)$; Eq. 3.4 omits the squared discrimination factor present in Eq. 3.3. Printed Chapter 4 probability equations and the normalized cumulative-exponential Appendix

B implementation differ. The implemented probabilities sum to one, but that fact is not faithful reproduction of the literal printed formula. These analytical findings are detailed in handoff/narrow-analytical-evidence.json.

Appendix G has probabilistic observation links. The binary link is $P(X = 1 \mid L) = 0.001 + 0.998\,1\{L \geq 0\}$, not a deterministic threshold. For a nondegenerate Gaussian liability it gives $0.001 + 0.998\Phi(\mu/\sigma)$ and scales binary cross-covariance by 0.998^2 under conditional independence. The continuous school-size observation has precision 100, adding variance 0.01 on its standardized scale. The regression uses observed or imputed covariates rather than simply their latent expectations. Thus the mixed-data application is not identical to the continuous joint-Gaussian simulation. See handoff/chapter6-analytical-evidence.json.

Chapter 7's software contribution is assessed separately from its empirical illustrations. BayesTwin version 1.0 is dated 2017; the available pre-defense revision is from 15 October 2015. Its example posterior data are simulated, not the dissertation's original empirical data (Schwabe, 2017). Version-aware software replay and probability-kernel checks establish only those specific implementation properties.

7 Assumptions

Our reconstruction needs explicit choices where the source does not uniquely specify an executable procedure. The following assumptions are supplied by the replication; source-stated Gaussian covariance structure and genetic correlations are model premises rather than newly inferred facts.

7.1 Low-missingness design

We use 2% as the primary low-missingness level and 3% as a separate sensitivity level because both appear in the source. Conclusions do not silently merge these designs.

7.2 Response-generator calibration and reliability summaries

We declare a fixed reconstructed calibration and reproducible seed rule. Raw and standardized reliability, participant subsets and across-dataset aggregation are alternative descriptors, not recovered historical choices. For Chapter 5, Bernoulli moderator probability 0.5 is consistent with approximately 11% explained variance but is not explicitly supplied by the source.

7.3 Sufficient conditions for identification

Known finite positive logistic calibrations, distinct conditionally independent observations, accessible joint moments and appropriate shared moderator/zygosity support are added sufficient mathematical hypotheses. They have not been empirically verified for the original applications. The common scale cannot be replaced by arbitrary nonlinear transformations.

7.4 Comparison margins and uncertainty

Source-coordinate bounds use 0.25 PDF point transformed through the source axes and summed across a linear contrast. Those bounds predate the guarded production experiment, although the full plan was not wholly preregistered and earlier outcomes were known. They bound extraction, not scientific

equivalence or source Monte Carlo error. Printed scalar slopes instead use half their last printed unit. No pooled margin is invented for multicomponent propositions or qualitative adequacy.

8 Ambiguities that were resolved or retained

The source contains distinctions that materially affect interpretation. An inconsistency is not automatically a replication failure; the following checks identify what can be resolved and what must stay uncertain.

8.1 A 01: Chapter 2 p 21 Eq 2.16 worked example

For Delta 1.5 prose multiplies .5×1.5=.75 though environmental baseline was .2; slope/ratio formula and baseline example conflict. Use the variance-ratio formula and retain the conflicting baseline example; the arithmetic discrepancy does not establish a posterior recovery result.

8.2 A 02: Chapter 3 p 37 Eq 3.4

OPLM information formula omits discrimination squared although Eq 3.3 has item discrimination; plot implementation unavailable from dissertation. The logistic derivative requires squared discrimination in information. The literal printed equation and model-implied information are distinct; the unavailable plot implementation is not inferred.

8.3 A 03: Chapter 3 p 41 versus p 42

Text defines effect at $A = \sigma_A^2$, but reported ratios 1.63/1.85 agree with one standard deviation. The reported ratios agree with one standard deviation rather than one variance unit; these are arithmetic checks on source parameters, not empirical re-estimation.

8.4 A 04: Chapter 4 p 56 Eqs 4.6–4.10 versus Appendix B

Printed GPCM normalization/threshold equations differ from cumulative-exponential code. The Appendix B cumulative-exponential implementation normalizes; the literal displayed expression does not generally define the same probability kernel.

8.5 A 05: Chapter 4 p 64

IRT h^2 mean .54 is outside printed 95% HPD [.34,.51]. The reported mean and HPD are retained without inferring a corrected posterior summary. A posterior mean can lie outside a highest-posterior-density interval; this alone does not establish an inconsistency.

8.6 A 06: Chapter 5 p 78 versus Table 5.1

Design prose says 80000 pairs but table and results say 8000. Preserve both; no silent replacement. The reconstruction uses the table/results value of 8,000 pairs and flags the 80,000 prose value. Neither choice recovers missing historical simulation records.

8.7 A 07: Chapter 5 p 74 versus p 84

Separate-moderator script referred to as Appendix F on p 74 but is Appendix E on p 84 and in contents. Appendix E contains the separate-moderator implementation; the Appendix F reference is a locator discrepancy.

8.8 A 08: Chapter 6 pp 91–95,101–102; Figs 6.3–6.6

Methods/captions say 2% at lowest missingness; plot labels/results/recommendation say 3%. Both levels are retained as distinct primary and sensitivity designs. The source-specific low-missingness comparison remains ambiguous.

8.9 A 09: Chapter 6 p 97 versus Table 6.1

Text says five pedagogical categories but enumerates six, which table confirms; Table 6.3 calls Reformed liberated Reformed liberal. The enumerated six categories and labels are preserved, with the wording discrepancy recorded; no independent application dataset is inferred.

8.10 A 10: Chapter 7 p 104 and Dutch summary p 117

Summaries list 2,3,4 covariates; primary Chapter 6 includes 5 as well. Do not omit q 5. All four primary covariate counts, including five, are retained. Shortened summary wording does not remove that design.

8.11 A 11: Dutch summary p 116 versus Chapter 4 abstract/results

Dutch wording says shared variance lower for genetic liberalism, opposite to primary Chapter 4 positive AxC result. Primary tables retained; translation discrepancy not treated as an independent experiment. The primary positive shared-environment slope is retained alongside the translation discrepancy; the summary is not an independent experiment.

8.12 A 12: Chapter 6 p 97 versus Appendix G pp 153–154

Prose deterministic threshold differs from .999/.001 binary link; program adds .01 continuous observation variance. The exact 0.999/0.001 binary kernel and 0.01 continuous observation variance are verified. Neither is silently replaced by deterministic observations.

8.13 A 13: Chapter 5 p 72 and Appendix C

Source equates unique model identification with a single finite-sample likelihood maximum; binary-reflection and continuous-moderator claims need different hypotheses. Population injectivity and finite-sample maximum existence/uniqueness are separate propositions. Binary reflection and rich continuous support require distinct hypotheses.

9 Availability of original inputs

The acquisition audit recovered the dissertation and model programs, but not the original empirical datasets or complete historical simulation environment. This is a documented search boundary, not proof that the missing inputs do not exist. The dated audit below is copied from `handoff/scientific-availability.json`; the original full dissertation and unlicensed programs are separate local inputs, not bundled reproduction files.

Institutional final-version record 2026-10-30. Verified PhD, institution, advisers, date, DOI, ISBN and final-version designation. Source location.

Author dissertation GitHub repository 2026-10-30. Full 164-page thesis and five model programs obtained at 6 c 2 c 48901209 fc 76 ba 932 bb 094 c 61 abba 5 ef 160 d. No raw empirical datasets or Monte Carlo driver/seeds. Full text read. Original files kept local; rights reserved/no code license. Source location.

All 12 public author GitHub repositories 2026-10-30. All cloned; immutable hashes in author-repositories.json. Data-file and code audit found workshop data, BayesTwin simulated posterior example and unrelated projects; no matching NTR/Cito/Virginia study inputs or curvilinear simulation driver. This is a search limit, not proof of absence. Source location.

OSF indexed search 2026-10-30. Queries site:osf.io Schwabe twin and exact Inga Schwabe 2016 found later unrelated projects/preprints, no matching dissertation simulation or empirical input. No identified thesis OSF project. Source location.

GitLab indexed search 2026-10-30. Queries Schwabe and exact ingaschwabe returned unrelated authors/no matching repository. No claim of exhaustive private-site search. Source location.

Zenodo indexed search 2026-10-30. Queries Schwabe twin and exact Inga Schwabe returned unrelated records/later projects; no matching dissertation data/code record. Source location.

Chapter 4 article and supplement 2026-10-30. Publisher confirms CC BY 4.0 article and links ten-page supplement. Supplement accessible through browser; contains category plots, fit plots and JAGS program, not raw responses. Original thesis code replayed locally. Supplement findings cannot be refitted without Virginia30K responses. Source location.

BayesTwin repository history 2026-10-30. Current code 656 afd 72 d 45 ecd 13 f 54 b 72 c 9400 cceface 8 d 423 f, version 1.0 dated 2017, GPL¿=2. Fetched full history; last pre-defense revision `a8e5d6b5ee6a8c76bdd0234a00fbcc809712ac08` dated 2015-10-15 available for separate replay. Example results.rda explicitly simulated, not dissertation empirical data. Source location.

Historical statistical engine specification 2026-10-30. Thesis cites R/rjags 2013 and OpenMx 2011/JAGS 2003 publications but no complete sessionInfo, engine patch versions, random seeds or driver. Exact historical environment not determined. Current pinned versions and original-model replay used with explicit scope limit. Source location.

10 Limits and unanswered questions

The evidence supports a partial replication under declared reconstruction assumptions. The following fourteen limits apply jointly; a completed calculation in one chapter does not close an empirical or posterior target elsewhere.

1. `H2.IRT-bias-remedy`, `H2.power-grid` and Chapter 5 Bayesian targets: the single fixed Chapter 2 floor case passes the extended operational diagnostic; repeated-dataset uncertainty and source 100/250-dataset posterior recovery/power remain unestablished. Other source designs and the distinct Chapter 5 model are not validated by this case.

2. Unavailable original empirical responses, calibration/linkage records, joint posterior draws and underreported designs prevent the corresponding independent empirical or exact-design comparisons.

3. Figure geometry and arithmetic from printed parameters reuse source evidence; they do not reproduce original data generation or empirical fits.

4. Conditional ordinary mathematical arguments do not supply formal certification or verify empirical calibration, observation and support assumptions.

5. `H6.precision-grid`: 36 primary conditions and 12 additional 3% missingness sensitivity conditions are distinct; low-missingness source 2% prose/3% plot ambiguity remains. Source-derived geometric bounds are neither original Monte Carlo uncertainty nor equivalence margins.

6. `C6-3.MAR-bias.A`, `C6-3.MAR-bias.C`, `C6-3.MAR-bias.h2`: 11 point sign reversals occur under full information; none excludes zero after adjustment. Both conditional MAR source-estimate deviations occur at ambiguous low missingness.

7. `H2.IRT-bias-remedy` and `H2.power-grid`: shorter original and centered gates failed, whereas the reviewed extended fixed-case gate passes. This does not complete any source 100/250-dataset posterior aggregate, power fraction or empirical effect.

8. `C1.scale-invariance` and Chapter 5/6 analytical claims: ordinary arguments are assumption-bound and not Lean-certified. Population identification is not finite-sample likelihood uniqueness.

9. Whole-dissertation inventory: 996 linked records include shared components, source-only arithmetic and parents; they are not 996 independent findings and no replication success fraction is defined.

10. `C2-2.design.alpha`, `C2-3.alpha`, `C5-2.alpha.J60` and `C5-2.alpha.J20`: source alpha formula/aggregation/calibration remain ambiguous; rounding of reconstructed means is not an exact source-estimand verdict. Standardization does not close the floor/ceiling or J 20 descriptor gaps.

11. `C6-2.SE-trend.MCAR.A` and other saved-target assessments: qualitative constancy, slightness, closeness and adequacy lack justified equivalence margins. Descriptive trends cannot settle those qualifications or `H6.power-assertion`.

12. `C5-1.logvariance-identification` and `C2-1.construction`: conditional ordinary proofs do not establish finite-sample maximum existence/uniqueness or observed-response identification of non-Gaussian latent moments.

13. `H2.power-grid` and `H2.IRT-bias-remedy`: all 12 whole/half-chain interaction HPDs include zero on this fixed null-interaction dataset; non-exclusion does not establish the null, unbiasedness or detection calibration. Equal-tailed quantile MCSE is not HPD endpoint MCSE. The identical shorter-run prefixes are dependent and are not pooled.

14. `H2.IRT-bias-remedy` and `H2.power-grid`: exact calibrated score likelihood is posterior-equivalent under fixed calibration, but its same-JAGS one-dataset pilot completed 0 of 4 prescribed 28000-draw chains within the fixed budget, retaining 16000 per chain. Full diagnostics/posterior comparison are ineligible; acceleration is not established. Historical timing is unequal-work and confounded, not a causal speed comparison. No source-count target is completed.

11 Conclusion

The reconstructed Chapter 6 experiment supports the direction of the full-information precision advantage and the deletion-bias pattern, while leaving literal precision magnitudes and the direct power assertion unresolved. Conditional mathematics clarifies scale and identification claims without proving a unique finite-sample likelihood maximum. Empirical interaction findings remain blocked by unavailable original inputs. Response generation and the completed single-case posterior diagnostic do not supply repeated-dataset recovery, and the bounded exact-score implementation did not complete its prescribed workload. Whole-dissertation replication is incomplete.

A Headline definitions and source comparisons

The first-page table contains every principal headline, plus two numerical illustrations selected independently of their displayed magnitudes. A nonscalar parent can be partly supported while its scalar magnitude is inapplicable. No average replaces its component evidence. Published values are from Schwabe (2016), with the specific locators below; our values are reconstructed results or explicitly conditional analytical findings.

A.1 HE 01 Common-scale item subsets

Under a common appropriately identified IRT measurement scale, biometric conclusions are asserted to be independent of the particular item subset. Source: Chapter 1 p 5; Chapter 4 p 51; Chapter 7 pp 107–108.

Assessment: partly replicated. A conditional Gaussian population-target version is supported by an independently reviewed ordinary argument. Literal finite-likelihood invariance does not follow. This does not assert that the dissertation intended exact finite-sample equality, and does not settle its non-Gaussian latent-genetic interaction models.

Comparison applicability: Conditional invariance under a common identified measurement scale is a theoretical relation across item sets, not one signed scalar effect. The proposition and its assumptions must be judged separately from any numeric magnitude. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No statistical interval applies to the conditional written argument. Exact rational covariance identities and strict derivative signs establish the stated restricted comparison; empirical assumptions and broader models are not validated by it.

A.2 HE 02 IRT recovery across designs

Incorporating an IRT measurement model remedies the scale-induced interaction bias and recovers generating parameters in the studied designs. Source: Chapter 2 abstract p 11; aim p 24; discussion p 25.

Assessment: not established. The completed extended diagnostic addresses one floor dataset only. The source 100-dataset parameter/scenario summaries, ceiling recovery and power calibration remain unresolved.

Comparison applicability: The remedy and recovery conclusion covers multiple parameters, units and scenarios, including unreported ceiling values. No scientifically justified scalar aggregate is published. Retain all component-specific source uncertainty and missing uncertainty; a qualitative conjunction has no invented pooled standard error. Constituents share model, generated data and sampling assumptions. No pooled uncertainty; source Monte Carlo and convergence gaps remain. One-dataset credible intervals are not across-replication uncertainty.

A.3 HE 03 Interaction power grid

The proposed method has useful power to detect genotype–environment interaction under the investigated sample-size and effect-size conditions. Source: Chapter 2 abstract p 11; results p 22; discussion p 25.

Assessment: not established. Short pilots are not 100-dataset detection fractions; no source-count posterior power grid is complete. The curvilinear extension design is unavailable.

Comparison applicability: The detection-performance conclusion concerns a grid of sample sizes and interaction effect sizes. All original detection rates and signed comparisons remain linked; no single grid average or selected cell is the whole headline. Retain all component-specific source uncertainty and missing uncertainty; a qualitative conjunction has no invented pooled standard error. Constituents share model, generated data and sampling assumptions. No pooled uncertainty; source Monte Carlo and convergence gaps remain. One-dataset credible intervals are not across-replication uncertainty.

A.4 HE 04 Sum-score scale artifacts

Sum-score analysis can produce spurious genotype–environment interaction from measurement-scale properties. Source: Chapter 2 abstract p 11; results pp 21–22; discussion pp 24–25.

Assessment: not established. Historical source-length one-dataset floor/ceiling/symmetric sumscore beta 1 means 1.0951643,-.8566462,.1003354 supply diagnostic patterns only. Source 100-dataset full constituent summaries and convergence are incomplete; no pooled artifact magnitude or component match fraction used.

Comparison applicability: Scale-induced spurious interaction has opposite signed floor and ceiling examples plus a symmetric control. No unique scalar can represent this multi-condition proposition; preserve each signed component. Retain all component-specific source uncertainty and missing uncertainty; a qualitative conjunction has no invented pooled standard error. Constituents share model, generated data and sampling assumptions. No pooled uncertainty; source Monte Carlo and convergence gaps remain. One-dataset credible intervals are not across-replication uncertainty.

A.5 HE 05 Mathematics: AxE slope

Unique-environmental variance increases with genetic mathematics ability; the AxE 95% HPD excludes zero. Source: Chapter 3 p 42, Table 3.2.

Assessment: blocked. Independent target C3-4.3.2.beta1 requires original NTR/Cito/SES responses, linkage/calibration or posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: Half the last printed decimal unit; descriptive point estimates, not equivalence. Published signed value 2.0837; ours not established. Units: log-variance slope per genetic-score unit. Posterior SD=0.4939; 95% HPD=[1.0886,3.0344]. This is not Monte Carlo uncertainty or evidence of equivalence. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.6 HE 06 Mathematics: AxC slope

The AxC point estimate is positive but its 95% HPD includes zero; shared-environment moderation is not detected. Source: Chapter 3 p 42, Table 3.2.

Assessment: blocked. Independent target C3-4.3.2.gamma1 requires original NTR/Cito/SES responses, linkage/calibration or posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: Half the last printed decimal unit; descriptive point estimates, not equivalence. Published signed value 2.6159; ours not established. Units: log-variance slope per genetic-score unit. Posterior SD=1.3503; 95% HPD=[-0.1358,5.2189]. This is not Monte Carlo uncertainty or evidence of equivalence. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.7 HE 07 Response dimensions

The 28-item scale measures two distinct dimensions: liberalism–conservatism and use of the question-mark response. Source: Chapter 4 abstract p 47, pp 58–61, Figs 4.1–4.4.

Assessment: blocked. Independent target C4-2.two-dimensions requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: The principal finding is the substantive two-dimensional interpretation of response behavior. The number two is a model-description count, not a signed scalar scientific contrast. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.8 HE 08 Social-liberalism subscale

Nine selected items constitute the proposed ordinal social-liberalism subscale. Source: Chapter 4 pp 62,66.

Assessment: blocked. Independent target C4-3.subscale requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: Construction and psychometric interpretation of the nine-item social-liberalism scale is a multicomponent methodological result. Item count nine is not an effect magnitude or a scalar adequacy contrast. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.9 HE 09 Liberalism: AxE slope

Unique-environmental variance decreases with genetic liberalism; the negative AxE 95% HPD excludes zero. Source: Chapter 4 p 63, Table 4.3.

Assessment: blocked. Independent target C4-5.4.3.beta1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: Half the last printed decimal unit; descriptive point estimates, not equivalence. Published signed value -2.81; ours not established. Units: log-variance slope per genetic-score unit. Posterior SD=0.21; 95% HPD=[-3.22,-2.39]. This is not Monte Carlo uncertainty or evidence of equivalence. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.10 HE 10 Liberalism: AxC slope

Shared-environmental variance increases with genetic liberalism; the positive AxC 95% HPD excludes zero. Source: Chapter 4 p 63, Table 4.3.

Assessment: blocked. Independent target C4-5.4.3.gamma1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: Half the last printed decimal unit; descriptive point estimates, not equivalence. Published signed value 0.54; ours not established. Units: log-variance slope per genetic-score unit. Posterior SD=0.14; 95% HPD=[0.31,0.84]. This is not Monte Carlo uncertainty or evidence of equivalence. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.11 HE 11 AxE scale reversal

The interaction point-estimate direction changes between IRT and sum-score analyses: β_1. Source: Chapter 4 pp 63–64,66–67, Tables 4.3–4.4.

Assessment: blocked. Independent target C4-5.reversal.beta1 requires Virginia30K responses and fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: The finding concerns the relationship between two signed AxE slopes estimated on different scales. Both slopes and their individual rounding tolerances remain available, but neither averaging nor subtraction of unstandardized cross-scale slopes represents this sign-reversal proposition. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.12 HE 12 AxC scale reversal

The interaction point-estimate direction changes between IRT and sum-score analyses: γ_1. Source: Chapter 4 pp 63–64,66–67, Tables 4.3–4.4.

Assessment: blocked. Independent target C4-5.reversal.gamma1 requires Virginia30K responses and

fitted/posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: The finding concerns the relationship between two signed AxC slopes estimated on different scales. Retain both scalar comparisons individually; no scientifically justified single aggregate expresses the reversal. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.13 HE 13 Log-variance identification

The alternative logvariance ACE×M model is asserted to be uniquely identified. Source: Chapter 5 pp 69,72–77,83–84; Appendix C pp 138–139.

Assessment: partly replicated. Conditional theorem only; empirical calibration/observation/support assumptions unverified and faithful Lean bridge absent.

Comparison applicability: Population model identification is an injectivity proposition conditional on model and observation assumptions, not a scalar effect size. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. A conditional exact ordinary argument under stated calibration, Gaussian model and observation-support assumptions; no sampling interval applies. Empirical validity of those assumptions and formal certification remain separate.

A.14 HE 14 Measurement-bias prevention

Simultaneous IRT and ACE×M modelling is asserted to prevent spurious moderation arising from heterogeneous measurement error. Source: Chapter 5 pp 69,72–77,83–84; Appendix C pp 138–139.

Assessment: not established. A general prevention claim is not established by single-dataset Chapter 2 diagnostics or conditional identification; further model/design checks remain feasible.

Comparison applicability: The principal method-level prevention-of-spurious-moderation claim ranges over measurement and biometric assumptions; no unique scalar bias reduction is published for this Chapter 5 assertion. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. Target uncertainty unestimated; no inference from adjacent diagnostics.

A.15 HE 15 Unique finite likelihood maximum

The source states the logvariance model has only one maximum in its likelihood function. Source: Chapter 5 pp 69,72–77,83–84; Appendix C pp 138–139.

Assessment: not established. Partially addressed; finite-sample at-most-one-maximum claim remains unresolved. Do not label it disproved solely by a no-finite-maximum example, and do not infer it from population injectivity.

Comparison applicability: Existence and uniqueness of a sample-likelihood maximum is a mathematical proposition, distinct from identification and from a numerical effect magnitude. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. Exact ordinary mathematical argument under the stated assumptions; no sampling interval or empirical calibration guarantee is implied. Numerical quadrature illustrates the nonexistence example but is not its proof.

A.16 HE 16 Sample size: A moderation

Large samples are said to be necessary to detect β_{1a} moderation. Source: Chapter 5 abstract p 69, pp 78,83.

Assessment: not established. No calibrated nonzero-effect power/adequacy experiment, decision criterion or justified adequacy margin has been run; null-interaction short pilots cannot establish detection or 20-item sufficiency. Feasible assessment remains pending.

Comparison applicability: The need for large samples to detect genetic moderation is an adequacy assertion inferred from uncertainty under null effects. No nonzero alternative, sufficient-power threshold or uniquely determined required sample size is reported, so it is not a single scalar power contrast. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No 250-dataset means/SDs or valid recovery uncertainty. Pilot ESS values are low and cannot establish convergence.

A.17 HE 17 Sample size: C moderation

Large samples are said to be necessary to detect β_{1c} moderation. Source: Chapter 5 abstract p 69, pp 78,83.

Assessment: not established. No calibrated nonzero-effect power/adequacy experiment, decision criterion or justified adequacy margin has been run; null-interaction short pilots cannot establish detection or 20-item sufficiency. Feasible assessment remains pending.

Comparison applicability: The need for large samples to detect shared-environment moderation is an adequacy assertion inferred from null-effect uncertainty. No quantitative sufficient-power threshold or uniquely determined sample-size effect is supplied. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No 250-dataset means/SDs or valid recovery uncertainty. Pilot ESS values are low and cannot establish convergence.

A.18 HE 18 Twenty-item sufficiency

As few as 20 phenotypic items are asserted sufficient to find moderation effects. Source: Chapter 5 pp 80,83.

Assessment: not established. No calibrated nonzero-effect power/adequacy experiment, decision criterion or justified adequacy margin has been run; null-interaction short pilots cannot establish detection or 20-item sufficiency. Feasible assessment remains pending.

Comparison applicability: Twenty items is a design condition, not the magnitude of the claim that this condition suffices for detecting interactions. Source adequacy lacks a specified power threshold and alternative; retain the full item-count evidence without inventing an effect. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. No 250-dataset means/SDs or valid recovery uncertainty. Pilot ESS values are low and cannot establish convergence.

A.19 HE 19 SES: C moderation

Shared-environmental variance is lower for high-SES families; the negative C×SES 95% HPD excludes zero. Source: Chapter 5 p 82, Table 5.4.

Assessment: blocked. Independent target C5-5.5.4.beta1c requires original NTR/Cito/SES responses,

linkage/calibration or posterior inputs absent after documented public-source acquisition. Printed numbers or source-only arithmetic are not independent empirical results.

Comparison applicability: Half the last printed decimal unit; descriptive point estimates, not equivalence. Published signed value -3.0777; ours not established. Units: log shared-environmental-variance ratio high versus low/average SES. Posterior SD=1.928; 95% HPD=[-7.3886,-0.03]. This is not Monte Carlo uncertainty or evidence of equivalence. No empirical sampling uncertainty or reproduction interval estimated for unavailable source target.

A.20 HE 20 Thirteen covariate nondetections

None of the thirteen measured covariate coefficients was detected in the educational-achievement application using the reported 99.6% HPD criterion. Source: Chapter 6 abstract p 86; results p 99 and Table 6.3 p 100; discussion p 102.

Assessment: blocked. The original linked NTR/Cito educational-achievement data and original fitted-model inputs were unavailable after documented permitted acquisition. Independent empirical counts, percentages, fitted estimates and nondetection results cannot be recovered from the Chapter 6 simulations.

Comparison applicability: The principal application finding is a conjunction of thirteen individual coefficient nondetections with different reference contrasts. No joint test or scalar average is reported. Retain every coefficient and interval; nondetection is not equivalence or zero effect. Retain all component-specific source uncertainty and missing uncertainty; a qualitative conjunction has no invented pooled standard error. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

A.21 HE 21 Full-information power

The source asserts that full information is more powerful than deletion regardless of the investigated missingness mechanism, covariate count or amount of missingness. Source: Chapter 6 abstract pp 85–86; pp 91,101–102; Chapter 7 p 104.

Assessment: not established. Lower estimated SE is not rejection-rate power. Chapter 6 provides no unique nonzero alternative, test/rejection rule or source power fractions. This production grid estimates precision and bias, not a direct power experiment; no new test definition is invented.

Comparison applicability: The source-wide greater-power assertion spans three mechanisms and a design grid and has no justified scalar aggregate. Its three retained scalar power-difference contrasts are separately classified scalar_unestablished because the source supplies no power precision or defensible comparison margin, not because replication results are missing. Retain all component-specific source uncertainty and missing uncertainty; a qualitative conjunction has no invented pooled standard error. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

A.22 HE 22 Full-information precision grid

Full information gives lower variance-component standard errors than deletion across the investigated missingness mechanisms and design grid. Source: Chapter 6 results pp 93–95; discussion p 101.

Assessment: partly replicated. All 108 signed deletion-minus-full variance-SE differences are positive and all 108 Bonferroni-adjusted intervals exclude zero positively. The precision ordering is reproduced

in the complete primary grid. Published literal variance-SE points differ from our estimates in all 216 conditional comparison bands; scientific comparability remains unresolved because original SE implementation and original Monte Carlo uncertainty are unavailable. Path SE is an exploratory different estimand. This does not establish rejection-rate power.

Comparison applicability: The principal precision conclusion spans 108 signed SE contrasts across three mechanisms, three variance components, four covariate counts and three missingness levels. All 108 parent contrasts and their existing extraction bounds remain; no pooled precision gain is invented. Retain all component-specific source uncertainty and missing uncertainty; a qualitative conjunction has no invented pooled standard error. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

A.23 HE 23 Observed-data likelihood

The full-information model uses all observed phenotype and covariate components when covariate data are incomplete. Source: Chapter 6 pp 88–91; Appendix F.

Assessment: partly replicated. The specified Gaussian construction and computational realization are supported. This restricted model-level result does not establish correct population distribution, empirical data validity or universal inference improvements.

Comparison applicability: Using all observed phenotype and covariate components is a likelihood-construction property, not a scalar effect. Its operational validity must be adjudicated independently of precision or power. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

A.24 HE 24 MAR: A bias pattern

MAR deletion underestimates A variance while full information remains close to the generating value. Source: Chapter 6 pp 94–95, Fig 6.5.

Assessment: partly replicated. Deletion bias is negative in 12/12 conditions, with 10/12 adjusted intervals excluding 0 in that direction; bias range [-0.0728311524251116,-0.0062175646058198804]. Full-information bias range [-0.0031917638748272703,0.0013358851221280952], with 0/12 adjusted intervals excluding 0. Thus deletion-bias direction is reproduced descriptively and supported in the stated subset. The full-information close-to-truth/unbiasedness component has no prespecified equivalence margin and is not established by intervals containing 0.

Comparison applicability: The MAR bias conclusion for A spans 12 design cells and two methods, retaining all 24 signed contrasts. Each contrast has its existing source extraction bound; no single scalar aggregate represents the complete bias pattern. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

A.25 HE 25 MAR: C bias pattern

MAR deletion underestimates C variance while full information remains close to the generating value. Source: Chapter 6 pp 94–95, Fig 6.5.

Assessment: partly replicated. Deletion bias is negative in 12/12 conditions, with 12/12 adjusted intervals excluding 0 in that direction; bias range [-0.11495778041479585,-0.007668582657040717]. Full-

information bias range [-0.002616392569283762,0.0010171452728346302], with 0/12 adjusted intervals excluding 0. Thus deletion-bias direction is reproduced descriptively and supported in the stated subset. The full-information close-to-truth/unbiasedness component has no prespecified equivalence margin and is not established by intervals containing 0.

Comparison applicability: The MAR bias conclusion for C spans 12 design cells and two methods, retaining all 24 signed contrasts. Each contrast has its existing source extraction bound; no single scalar aggregate represents the complete bias pattern. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

A.26 HE 26 MAR: heritability bias pattern

MAR deletion overestimates heritability while full information remains close to the generating value. Source: Chapter 6 pp 94–95, Fig 6.5.

Assessment: partly replicated. Deletion bias is positive in 12/12 conditions, with 9/12 adjusted intervals excluding 0 in that direction; bias range [0.0014162153151865 44,0.031180388166267718]. Full-information bias range [-0.001390288356138391,0.0026536390348624117], with 0/12 adjusted intervals excluding 0. Thus deletion-bias direction is reproduced descriptively and supported in the stated subset. The full-information close-to-truth/unbiasedness component has no prespecified equivalence margin and is not established by intervals containing 0.

Comparison applicability: The MAR bias conclusion for h 2 spans 12 design cells and two methods, retaining all 24 signed contrasts. Each contrast has its existing source extraction bound; no single scalar aggregate represents the complete bias pattern. No uncertainty estimate is published for this particular quantity; source numerical rounding is separate from uncertainty. Each numeric component retains its own Monte Carlo and extraction uncertainty; no pooled effect or pooled uncertainty is invented.

B Complete component evidence and inventory coverage

All 996 original records are mapped to their scientific chapter or appendix below and retained, without dropping unresolved targets, in `coverage-v4.csv` and `handoff/a0-evidence-packet.json`. The inventory's exact-target work counts are 255 blocked, 391 pending, 29 attempted-incomplete, 11 completed, and 310 completed under the declared reconstruction. These are work-scope counts, not scientific verdict totals or a success fraction. Source-only arithmetic and shared parent/component records cannot be counted as independent experiments.

The 187 signed headline components are retained in `paper/a0-component-comparisons.json`, a lossless extraction of each parent's comparison records with exact definitions, source values, our values, tolerances and uncertainty. The table below prints their scalar values and point labels without inventing a grid-level scalar. For the 108 precision components, magnitude is unresolved even where a conditional arithmetic label could be computed. For MAR bias components, extraction-based labels are descriptive; source Monte Carlo uncertainty is unknown. For the three power contrasts, the source reports a full-information power advantage but supplies no quantified power differences. Their Published cells therefore say not numerically reported; our direct power comparison is not established. These contrasts have no defensible source precision or prespecified margin. These cannot be tested by substituting precision.

Source section	Coverage and scientific discussion	Records
chapter 1	PDF pages 10–18; printed pages 1–9	7
chapter 2	PDF pages 20–36; printed pages 11–27	223
chapter 3	PDF pages 38–55; printed pages 29–46	54
chapter 4	PDF pages 56–77; printed pages 47–68	98
chapter 5	PDF pages 78–93; printed pages 69–84	206
chapter 6	PDF pages 94–111; printed pages 85–102	408
chapter 7	PDF pages 112–123; printed pages 103–114	8
appendix A	PDF pages 140–142; printed pages 131–133	3
appendix B	PDF pages 143–146; printed pages 134–137	8
appendix C	PDF pages 147–148; printed pages 138–139	4
appendix D	PDF pages 149–151; printed pages 140–142	1
appendix E	PDF pages 152–155; printed pages 143–146	2
appendix F	PDF pages 156–158; printed pages 147–149	2
appendix G	PDF pages 159–163; printed pages 150–154	3
frontmatter	PDF pages 1–9. Degree/title/edition metadata and table of contents; no original scientific result.	0
dutch summary	PDF pages 124–126. Translated summaries duplicate mapped Chapter 2–6 claims; no new experiment. Wording discrepancies retained in ambiguities.	0
bibliography	PDF pages 128–136. Cited prior works are background; their scientific results are not original dissertation replication targets.	0
acknowledgements	PDF pages 138–138. Acknowledgements of collaborators/funding, not original scientific results.	0
backcover	PDF pages 164–164. Print production marks, not a scientific result.	0

B.1 Signed component table

Each row key gives its parent headline and zero-based contrast index. The exact scientific definition and units are in the linked component file. HE11–12 retain separately scaled interaction slopes; HE21 retains missing power differences; HE22 is deletion minus full-information variance SE; HE24–26 are estimate-minus-generating-value biases for A, C and heritability. Positive HE22 means full information has lower SE; positive HE24–26 means upward bias. Row-specific extraction tolerances apply to magnitude arithmetic, not inference. NE in Our finding, magnitude or direction means not established by this replication; it does not describe the source claim. S means same direction; R means reversed direction; Z means zero versus nonzero.

Parent	Index	Published	Ours	Margin	Magnitude	Direction
HE11	0	-2.81000000	NE	0.00500000	NE	NE
HE11	1	0.10000000	NE	0.00500000	NE	NE
HE12	0	0.54000000	NE	0.00500000	NE	NE
HE12	1	-2.21000000	NE	0.00500000	NE	NE
HE21	0	not numerically reported	NE	NE	NE	NE
HE21	1	not numerically reported	NE	NE	NE	NE
HE21	2	not numerically reported	NE	NE	NE	NE
HE22	0	0.00106356	0.00150235	0.00023667	NE	S
HE22	1	0.00356296	0.00490662	0.00023667	NE	S
HE22	2	0.00654293	0.00911630	0.00023667	NE	S
HE22	3	0.00194661	0.00276225	0.00023667	NE	S
HE22	4	0.00658042	0.00922015	0.00023667	NE	S
HE22	5	0.01236273	0.01701897	0.00023667	NE	S
HE22	6	0.00282129	0.00408378	0.00023667	NE	S
HE22	7	0.00991674	0.01383328	0.00023667	NE	S
HE22	8	0.01880491	0.02593898	0.00023667	NE	S
HE22	9	0.00390503	0.00540764	0.00023667	NE	S
HE22	10	0.01343646	0.01881441	0.00023667	NE	S
HE22	11	0.02703050	0.03632223	0.00023667	NE	S
HE22	12	0.00140397	0.00142765	0.00027646	NE	S
HE22	13	0.00442406	0.00462077	0.00027646	NE	S
HE22	14	0.00831621	0.00843680	0.00027646	NE	S
HE22	15	0.00231474	0.00245377	0.00027646	NE	S
HE22	16	0.00777751	0.00818103	0.00027646	NE	S
HE22	17	0.01493045	0.01522838	0.00027646	NE	S
HE22	18	0.00349311	0.00356375	0.00027646	NE	S
HE22	19	0.01183299	0.01206663	0.00027646	NE	S
HE22	20	0.02264228	0.02266608	0.00027646	NE	S
HE22	21	0.00440404	0.00459668	0.00027646	NE	S
HE22	22	0.01577396	0.01611550	0.00027646	NE	S
HE22	23	0.03074147	0.03127694	0.00027646	NE	S
HE22	24	0.00038198	0.00033850	0.00015467	NE	S
HE22	25	0.00128720	0.00113664	0.00015467	NE	S
HE22	26	0.00273035	0.00219713	0.00015467	NE	S
HE22	27	0.00076970	0.00073041	0.00015467	NE	S
HE22	28	0.00263221	0.00241177	0.00015467	NE	S
HE22	29	0.00498241	0.00434776	0.00015467	NE	S
HE22	30	0.00118334	0.00110027	0.00015467	NE	S
HE22	31	0.00415224	0.00373305	0.00015467	NE	S
HE22	32	0.00791956	0.00710514	0.00015467	NE	S
HE22	33	0.00172214	0.00152493	0.00015467	NE	S
HE22	34	0.00593494	0.00532185	0.00015467	NE	S

Parent	Index	Published	Ours	Margin	Magnitude	Direction
HE22	35	0.01182568	0.01008825	0.00015467	NE	S
HE22	36	0.00106794	0.00124866	0.00015771	NE	S
HE22	37	0.00319499	0.00364190	0.00015771	NE	S
HE22	38	0.00541063	0.00566546	0.00015771	NE	S
HE22	39	0.00194052	0.00235636	0.00015771	NE	S
HE22	40	0.00557137	0.00637341	0.00015771	NE	S
HE22	41	0.00902209	0.00927196	0.00015771	NE	S
HE22	42	0.00274663	0.00336987	0.00015771	NE	S
HE22	43	0.00784172	0.00885134	0.00015771	NE	S
HE22	44	0.01236331	0.01258415	0.00015771	NE	S
HE22	45	0.00361450	0.00439310	0.00015771	NE	S
HE22	46	0.00997268	0.01081951	0.00015771	NE	S
HE22	47	0.01553050	0.01489491	0.00015771	NE	S
HE22	48	0.00182794	0.00099990	0.00031543	NE	S
HE22	49	0.00562604	0.00280105	0.00031543	NE	S
HE22	50	0.00990351	0.00417290	0.00031543	NE	S
HE22	51	0.00299204	0.00178769	0.00031543	NE	S
HE22	52	0.00944778	0.00467169	0.00031543	NE	S
HE22	53	0.01679175	0.00637724	0.00031543	NE	S
HE22	54	0.00404840	0.00251474	0.00031543	NE	S
HE22	55	0.01273575	0.00628138	0.00031543	NE	S
HE22	56	0.02283779	0.00830730	0.00031543	NE	S
HE22	57	0.00553384	0.00317116	0.00031543	NE	S
HE22	58	0.01670179	0.00747473	0.00031543	NE	S
HE22	59	0.03189262	0.00928477	0.00031543	NE	S
HE22	60	0.00036873	0.00032371	0.00011829	NE	S
HE22	61	0.00105491	0.00092911	0.00011829	NE	S
HE22	62	0.00174826	0.00145989	0.00011829	NE	S
HE22	63	0.00071796	0.00063713	0.00011829	NE	S
HE22	64	0.00203470	0.00178975	0.00011829	NE	S
HE22	65	0.00324186	0.00276531	0.00011829	NE	S
HE22	66	0.00109308	0.00095173	0.00011829	NE	S
HE22	67	0.00301325	0.00264961	0.00011829	NE	S
HE22	68	0.00462801	0.00388141	0.00011829	NE	S
HE22	69	0.00152248	0.00132513	0.00011829	NE	S
HE22	70	0.00401475	0.00331059	0.00011829	NE	S
HE22	71	0.00606742	0.00497513	0.00011829	NE	S
HE22	72	0.00082977	0.00109008	0.00022548	NE	S
HE22	73	0.00270859	0.00358782	0.00022548	NE	S
HE22	74	0.00495257	0.00640264	0.00022548	NE	S
HE22	75	0.00166820	0.00237452	0.00022548	NE	S
HE22	76	0.00548090	0.00755345	0.00022548	NE	S
HE22	77	0.00973640	0.01338838	0.00022548	NE	S

Parent	Index	Published	Ours	Margin	Magnitude	Direction
HE22	78	0.00261919	0.00365077	0.00022548	NE	S
HE22	79	0.00843331	0.01165845	0.00022548	NE	S
HE22	80	0.01460104	0.02072288	0.00022548	NE	S
HE22	81	0.00368420	0.00497937	0.00022548	NE	S
HE22	82	0.01161689	0.01592383	0.00022548	NE	S
HE22	83	0.02043990	0.02829504	0.00022548	NE	S
HE22	84	0.00115151	0.00113577	0.00022548	NE	S
HE22	85	0.00366410	0.00377388	0.00022548	NE	S
HE22	86	0.00651262	0.00664253	0.00022548	NE	S
HE22	87	0.00204781	0.00221637	0.00022548	NE	S
HE22	88	0.00661665	0.00708443	0.00022548	NE	S
HE22	89	0.01299785	0.01256046	0.00022548	NE	S
HE22	90	0.00309548	0.00328351	0.00022548	NE	S
HE22	91	0.01043490	0.01049286	0.00022548	NE	S
HE22	92	0.01982926	0.01856977	0.00022548	NE	S
HE22	93	0.00430912	0.00433691	0.00022548	NE	S
HE22	94	0.01390567	0.01401259	0.00022548	NE	S
HE22	95	0.02525541	0.02491638	0.00022548	NE	S
HE22	96	0.00027269	0.00017863	0.00022548	NE	S
HE22	97	0.00090630	0.00060187	0.00022548	NE	S
HE22	98	0.00167548	0.00114072	0.00022548	NE	S
HE22	99	0.00065222	0.00053424	0.00022548	NE	S
HE22	100	0.00215024	0.00172841	0.00022548	NE	S
HE22	101	0.00388469	0.00308362	0.00022548	NE	S
HE22	102	0.00108962	0.00092365	0.00022548	NE	S
HE22	103	0.00352839	0.00300368	0.00022548	NE	S
HE22	104	0.00616347	0.00541729	0.00022548	NE	S
HE22	105	0.00161347	0.00135791	0.00022548	NE	S
HE22	106	0.00514465	0.00426028	0.00022548	NE	S
HE22	107	0.00899626	0.00758239	0.00022548	NE	S
HE24	0	-0.00605332	-0.00621756	0.00059075	NE	S
HE24	1	-0.00288567	-0.00156831	0.00059075	NE	S
HE24	2	-0.01093086	-0.01374588	0.00059075	larger	S
HE24	3	0.00254044	0.00023544	0.00059075	smaller	S
HE24	4	-0.02759837	-0.02699159	0.00059075	smaller	S
HE24	5	-0.00173849	-0.00144257	0.00059075	same	S
HE24	6	-0.00590924	-0.00642497	0.00059075	NE	S
HE24	7	-0.00012927	-0.00082666	0.00059075	NE	S
HE24	8	-0.02173952	-0.02168936	0.00059075	same	S
HE24	9	0.00041893	-0.00003080	0.00059075	same	R
HE24	10	-0.04342901	-0.04647127	0.00059075	larger	S
HE24	11	-0.00245962	-0.00274859	0.00059075	same	S
HE24	12	-0.00752494	-0.00712869	0.00059075	NE	S

Parent	Index	Published	Ours	Margin	Magnitude	Direction
HE24	13	0.00077236	0.00045296	0.00059075	NE	S
HE24	14	-0.03178341	-0.03114017	0.00059075	smaller	S
HE24	15	-0.00076452	0.00004929	0.00059075	smaller	R
HE24	16	-0.05655388	-0.05309082	0.00059075	smaller	S
HE24	17	-0.00067063	0.00133589	0.00059075	larger	R
HE24	18	-0.00731596	-0.01474011	0.00059075	NE	S
HE24	19	0.00290101	-0.00319176	0.00059075	NE	R
HE24	20	-0.04070882	-0.03988467	0.00059075	smaller	S
HE24	21	-0.00143503	-0.00132062	0.00059075	same	S
HE24	22	-0.07050127	-0.07283115	0.00059075	larger	S
HE24	23	0.00306737	0.00044195	0.00059075	smaller	S
HE25	0	-0.00913151	-0.00766858	0.00066545	NE	S
HE25	1	-0.00054502	0.00010051	0.00066545	NE	R
HE25	2	-0.02936369	-0.02857453	0.00066545	smaller	S
HE25	3	-0.00261399	-0.00155888	0.00066545	smaller	S
HE25	4	-0.04861458	-0.04813777	0.00066545	same	S
HE25	5	-0.00103566	-0.00011525	0.00066545	smaller	S
HE25	6	-0.01508538	-0.01533751	0.00066545	NE	S
HE25	7	-0.00292494	-0.00176942	0.00066545	NE	S
HE25	8	-0.04162193	-0.04254259	0.00066545	larger	S
HE25	9	-0.00072446	-0.00093569	0.00066545	same	S
HE25	10	-0.07321303	-0.06918151	0.00066545	smaller	S
HE25	11	-0.00128894	0.00025119	0.00066545	smaller	R
HE25	12	-0.01775067	-0.01971251	0.00066545	NE	S
HE25	13	-0.00199224	-0.00200231	0.00066545	NE	S
HE25	14	-0.05260500	-0.05656576	0.00066545	larger	S
HE25	15	0.00041156	-0.00202670	0.00066545	larger	R
HE25	16	-0.09284027	-0.09768223	0.00066545	larger	S
HE25	17	-0.00119081	-0.00087975	0.00066545	same	S
HE25	18	-0.02784206	-0.02054156	0.00066545	NE	S
HE25	19	-0.00515686	0.00101715	0.00066545	NE	R
HE25	20	-0.06913269	-0.07073267	0.00066545	larger	S
HE25	21	-0.00074144	-0.00043337	0.00066545	same	S
HE25	22	-0.11634371	-0.11495778	0.00066545	smaller	S
HE25	23	-0.00324353	-0.00261639	0.00066545	same	S
HE26	0	0.00172809	0.00141622	0.00027219	NE	S
HE26	1	-0.00117418	-0.00031479	0.00027219	NE	S
HE26	2	0.01059974	0.00882709	0.00027219	smaller	S
HE26	3	0.00284939	0.00129508	0.00027219	smaller	S
HE26	4	0.01338730	0.01327260	0.00027219	same	S
HE26	5	0.00013556	-0.00029792	0.00027219	same	R
HE26	6	0.00532734	0.00554308	0.00027219	NE	S
HE26	7	0.00187285	0.00116057	0.00027219	NE	S

Parent	Index	Published	Ours	Margin	Magnitude	Direction
HE26	8	0.01249201	0.01262482	0.00027219	same	S
HE26	9	0.00129049	0.00087666	0.00027219	smaller	S
HE26	10	0.01919505	0.01551671	0.00027219	smaller	S
HE26	11	-0.00059174	-0.00105639	0.00027219	larger	S
HE26	12	0.00674144	0.00767138	0.00027219	NE	S
HE26	13	0.00248891	0.00197966	0.00027219	NE	S
HE26	14	0.01367013	0.01627894	0.00027219	larger	S
HE26	15	-0.00000213	0.00190514	0.00027219	larger	R
HE26	16	0.02494554	0.03061640	0.00027219	larger	S
HE26	17	0.00071111	0.00211800	0.00027219	larger	S
HE26	18	0.01252875	0.00410212	0.00027219	NE	S
HE26	19	0.00521278	-0.00139029	0.00027219	NE	R
HE26	20	0.01952200	0.02096849	0.00027219	larger	S
HE26	21	0.00066423	0.00068233	0.00027219	same	S
HE26	22	0.03315402	0.03118039	0.00027219	smaller	S
HE26	23	0.00413201	0.00265364	0.00027219	smaller	S

C Reproduction, software and artifacts

Rebuilding this manuscript from saved evidence is separate from regenerating datasets, refitting models or obtaining the original source data. Work from `formalization/schwabe-psychometric-genetics`. The editable manuscript is `paper/a0-standalone.tex`; its bibliography is `paper/references.bib`. Run `paper/build-a0.sh` to build `paper/a0-main.pdf`. The build uses the supplied preamble, pdfLaTeX and BibTeX; it does not execute a study.

Reproduction revision identifies the immutable scientific inputs and outputs, including the bounded score-model code, four incomplete chains, saved states and terminal resource records. It is a scientific snapshot; the manuscript rendering has its own hash-bound build record. The exact read-only hash verification recipe is `handoff/score-a0-reproduction.md`. Its expected result is zero completed fixed-length chains, 6419.751313023 summed chain/cleanup seconds and 1606.104005898 elapsed seconds. No sampling or full diagnostic computation is needed to verify these saved hashes.

The Gaussian experiment recipe is `handoff/terminal-reproduction.md`; the completed diagnostic and analytical-output recipe is `handoff/current-reproduction.md`. Source-text-dependent checks require the separately held dissertation with its recorded hash. Saved-output validation and deterministic algebra checks do not recreate original empirical data. A new sampling run is a distinct experiment; the incomplete score chains are not a basis for extending or pooling the recorded result.

R 4.5.0 supplies the statistical runtime (R Core Team, 2025). OpenMx 2.21.13 implements the Gaussian raw-data likelihood (Neale et al., 2016); lavaan 0.6.19 supplies baseline engine comparison (Rosseel, 2012). The item-level non-Gaussian hierarchy uses JAGS 4.3.2 through rjags 4-16 rather than a covariance-only approximation (Plummer, 2024). Posterior 1.6.1 supplies effective-sample-size and MCSE calculations; coda 0.19.4.1 supplies HPD summaries (Plummer et al., 2006). R package version and Debian packaging version are different identifiers. Python 3.12.14 performs saved-output and exact-arithmetic checks; no Lean or Mathlib formal certificate is supplied. Runtime identities are retained in `provenance/continuation-python-runtime.json`, `handoff/continuation-runtime-version-evidence.json` and the

reproduction records.

Software distribution links are R, OpenMx, lavaan, JAGS, rjags, posterior, coda and Python. These links identify tools; they do not establish the correctness of a reconstruction.

D Compliance ledger

The following ledger preserves the declared scope and departures of the scientific work. Completion of document checks is not scientific validation.

Whole-dissertation coverage – followed The source inventory covers 19 sections, 996 dependent source records, 26 headline claims and 187 effect components. Exact-target work comprises 255 blocked, 391 pending, 29 attempted-incomplete, 11 completed and 310 completed under the declared reconstruction (coverage-v4.csv). Inventory coverage does not establish whole-study replication.

Original-source and input audit – followed The input audit examined all 12 public author repositories. Original empirical responses, linked calibration and some simulation specifications were unavailable; their associated targets remain unresolved (handoff/scientific-availability.json).

Preferred statistical engines – followed R 4.5.0, lavaan 0.6.19 and OpenMx 2.21.13 were available. OpenMx implements the Chapter 6 Gaussian likelihood, with independent baseline engine checks recorded in outputs/engine-agreement.csv; package versions are recorded in provenance/preflight.json.

JAGS for non-Gaussian latent item models – departed A covariance-only SEM would change the source likelihood. JAGS 4.3.2 with rjags 4-16 reconstructs the item-level models; posterior 1.6.1 computes diagnostics. Generator, baseline engine and model-semantics checks do not independently validate the full posterior. The item-level posterior and its diagnostic limits are specified in handoff/jags-gate-scientific-evidence.json and handoff/jags-centered-scientific-evidence.json. The extended one-dataset result and runtime versions, including coda 0.19.4.1, are recorded in handoff/jags-centered-extended-recovery-scientific-evidence.json.

Original repetition counts and adequate posterior convergence – departed Chapter 6 completes 1000 datasets in 36 primary and 12 sensitivity conditions, totaling 96000 method fits. Response generation completes 2650 datasets across 19 Chapter 2/5 conditions; alpha sensitivity adds zero independent datasets and zero posterior fits. Shorter original and centered diagnostics failed their joint criteria. The completed four-chain centered diagnostic with 28000 retained draws per chain passes the operational criteria for all 14 monitored quantities on one fixed Chapter 2 N 500/J 60 floor dataset; its interaction HPD includes zero. This does not establish the source 100/250-dataset posterior recovery or power aggregates, other designs, or empirical refits. production/chapter6-v1/terminal-analysis/summary.json; handoff/generator-summary-scientific-evidence.json; handoff/generator-alpha-sensitivity-scientific-evidence.json; handoff/jags-centered-extended-recovery-scientific-evidence.json.

Signed estimands, uncertainty and multiplicity – followed The analysis distinguishes variance and path SEs, deletion-minus-full contrasts, estimate-minus-truth bias and source deviations (production/analysis-plan.json). Geometric extraction uncertainty, our Monte Carlo error and unknown source Monte Carlo error remain distinct. Descriptive magnitude is not equivalence.

Analytical verification and formal proof – departed Ordinary mathematical arguments and exact algebra checks support the explicitly conditional results; no Lean certificate is claimed. Population identification, fixed-parameter latent moment inversion and exact Rasch score-likelihood sufficiency do not establish finite-likelihood uniqueness, non-Gaussian observed-item identification, empirical recovery, power or convergence. The derivations and assumptions appear in handoff/chapter5-analytical-evidence.json, handoff/chapter6-analytical-evidence.json, provenance/continuous-moderator-identification-check.json, provenance/non-gaussian-latent-moment-check.json and provenance/rasch-sufficiency-derivation.json.

Source attribution and permitted redistribution – followed Schwabe (2016) is identified by verified bibliographic metadata and claim-specific dissertation locators. The full dissertation and unlicensed original programs remain local; redistributed materials comprise original replication code, generated results and limited attributed source extracts (handoff/scientific-availability.json).

Reproducible versions and commands – followed Immutable repository revision `2424e0f4d6 7929ba72255ef6f546ba2a72c25df0` contains completed scientific outputs, diagnostic draws, reviewed checks and original replication code. Saved-output validation and fixed-diagnostic rerun commands are specified in handoff/current-reproduction.md; handoff/terminal-reproduction.md supplies the Chapter 6 recipe. Full copyrighted dissertation text and unlicensed original programs remain separate inputs.

Equivalent score-likelihood implementation feasibility – departed The same-JAGS exact calibrated score representation was tested on one saved dataset with fixed starts and priors; it is not independent-engine validation. All four chains adapted and retained 16000 of 28000 requested draws before the preset summed-process ceiling; zero fixed-length chains completed. 6419.751313 chain/cleanup seconds and 6420.894927 all-phase seconds were consumed. Full joint diagnostic and five-global comparison were ineligible. Source-count targets remain pending. handoff/score-a0-scientific-evidence.json; provenance/score-a0-terminal-independent-review.json.

Manuscript layout – departed The approved article preamble is retained. The 28-row overview uses a compact 8-point type size to keep the full headline set on page 1; larger-type definitions and complete component evidence follow. Nonscalar rows preserve unavailable magnitudes separately from scientific assessments. The full inventory is provided as a source-linked artifact and chapter map rather than 996 repeated prose paragraphs.

References

Olive Jean Dunn. Multiple comparisons among means. *Journal of the American Statistical Association*, 56(293):52–64, 1961. doi: 10.1080/01621459.1961.10482090.

Michael C. Neale, Michael D. Hunter, Joshua N. Pritikin, Mahsa Zahery, Timothy R. Brick, Robert M. Kirkpatrick, Ryne Estabrook, Timothy C. Bates, Hermine H. Maes, and Steven M. Boker. Openmx 2.0: Extended structural equation and statistical modeling. *Psychometrika*, 81(2):535–549, 2016. doi: 10.1007/s11336-014-9435-8.

Martyn Plummer. *rjags: Bayesian Graphical Models using MCMC*, 2024. URL https://CRAN.R-project.org/package=rjags. R package version 4-16; used with JAGS 4.3.2.

Martyn Plummer, Nicky Best, Kate Cowles, and Karen Vines. Coda: Convergence diagnosis and output analysis for mcmc. *R News*, 6(1):7–11, 2006. URL https://www.r-project.org/doc/Rnews/Rnews_2006-1.pdf.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, 2025. URL https://www.R-project.org/. Version 4.5.0.

Yves Rosseel. lavaan: An r package for structural equation modeling. *Journal of Statistical Software*, 48 (2):1–36, 2012. doi: 10.18637/jss.v048.i02.

Inga Schwabe. *Nature, Nurture and Item Response Theory: A Psychometric Approach to Behaviour Genetics*. PhD thesis, University of Twente, 2016. URL https://research.utwente.nl/en/publications/nature-nurture-and-item-response-theory-a-psychometric-approach-t/.

Inga Schwabe. *BayesTwin: Bayesian Analysis of Item-Level Twin Data*, 2017. URL https://github.com/ingaschwabe/BayesTwin. Version 1.0; also inspected pre-defense repository revision dated 15 October 2015.

Inga Schwabe, Wilfried Jonker, and Stéphanie M. van den Berg. Genes, culture and conservatism—a psychometric-genetic approach. *Behavior Genetics*, 46:516–528, 2016. doi: 10.1007/s10519-015-9768-9.

Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of mcmc. arXiv:1903.08008, version 5, 22 June 2021, 2021. URL https://arxiv.org/abs/1903.08008v5.

Felix Voigtlaender. A general version of Price's theorem, 2020. URL https://arxiv.org/html/1710.03576v2. Version 2, June 1, 2020; Corollary 2.